Jan Škvrna

PhD Student in Computer Vision | 3D Perception & Spatial Understanding

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Selected Publications

IJCV 2026 MonoSOWA++: Universal Autolabeling Pipeline for Monocular 3D Object Detection. **J. Škvrna**, L. Neumann — accepted for publication. [Code]

ICCV 2025 MonoSOWA: Scalable Monocular 3D Object Detector Without Human Annotations. **J. Škvrna**, L. Neumann. [Paper] [Code]

ECCV 2024 TCC-Det: Temporarily Consistent Cues for Weakly-Supervised 3D Detection. **J. Škvrna**, L. Neumann. [Paper] [Code]

Research Experience

PhD Researcher — 3D Computer Vision | Visual Recognition Group, CTU Prague | Feb 2024 – Present

Training 3D detectors without target-domain human annotations

- **TCC-Det** (**ECCV** 2024) — auto-labelling pipeline that trains a LiDAR 3D detector with no target-domain human annotations, using an off-the-shelf 2D instance segmentation model, temporal consistency across frames, and a novel Template Fitting Loss based on a relaxed, outlier-robust variant of Chamfer distance. Set a new state of the art among annotation-free methods on KITTI and provided the first weakly-supervised evaluation on the Waymo Open Dataset.
- **MonoSOWA** (**ICCV** 2025) — extended the approach to monocular video without target-domain human annotations, separating object from camera motion and normalising focal length across datasets. KITTI-360 pseudo-label pre-training plus 25% of KITTI labels outperformed the same detector trained from scratch on 100%.
- **MonoSOWA++** (**IJCV**, accepted 2026) — unified both into one pipeline at 1.9 s/frame, an approximately 500x speed-up over the previous state of the art, adding confidence-scored pseudo-labels and support for pedestrians and cyclists.

Structured 3D reconstruction — S23DR Challenge, CVPR Urban Scene Modeling Workshop

Developed and implemented both winning methods end to end in teams of three (2026) and two (2025).

- **2026 — 1st place**. Flow-matching DiT over vertex sets with Perceiver-style scene conditioning. [Paper] [Code]
- **2025 — 1st place**. Native-3D, two-stage PointNet pipeline for vertex refinement and edge prediction. [Paper] [Code]

Awards & Honours

1st Place — **S23DR Challenge 2026** | Urban Scene Modeling Workshop, **CVPR 2026** — $12k prize pool | 2026

1st Place — **S23DR Challenge 2025** | Urban Scene Modeling Workshop, **CVPR 2025** — $25k prize pool | 2025

Dean's Award for Exceptional Master's Thesis | Czech Technical University in Prague | 2024

Education

PhD in Informatics | Czech Technical University in Prague | Feb 2024 – Dec 2027 (expected)

Visual Recognition Group (led by Prof. Jiří Matas), supervised by Dr. Lukáš Neumann.

MSc in Cybernetics and Robotics, with Honours | Czech Technical University in Prague | Sep 2021 – Feb 2024

Erasmus exchange, University of Southern Denmark (Sep 2022 – Feb 2023).

Industry & Teaching Experience

Junior Embedded Software Developer | Poll s.r.o. | Jun 2020 – Jun 2023

Real-time C/C++ firmware for ARM microcontrollers in HVAC and high-voltage DC/DC converter systems.

Lab Instructor — Deep Learning Essentials | CTU Prague | 2025 – Present

Academic Service & Training

Reviewing: ICCV 2025, BMVC 2026, AAAI 2027

Invited Talks: CMP Colloquium, Prague (2026) | Computer Vision Winter Workshop, Graz (2025)

Summer Schools: EEML, Montenegro (2026) | Int. Computer Vision Summer School (ICVSS), Sicily (2025)

Technical Skills

Languages & Systems: Python, C/C++, CUDA, Linux, Git, Slurm GPU clusters, distributed PyTorch training (DDP)

ML / Vision: PyTorch, OpenCV, Open3D, PyTorch3D — 3D detection & reconstruction, multi-view geometry, depth & pose estimation, weak/self-supervision, flow matching